

HYPERFINE STRUCTURES OF IRON(II) SULPHATES: MELANTERITE AND ROZENITE

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Mössbauer spectra of melanterite display two different iron environments. The difference in the quadrupole splittings comes from both the lattice and valence contributions. The lattice contributions for the two ferrous sites are of equal magnitude but of opposite signs while the valence contributions are both positive. Spectra of rozenite reveal only one ferrous site; a sample obtained from incomplete transformation of melanterite into rozenite shows that the low temperature quadrupole splitting of ferrous ions could be as high as 4 mm/s.

1. Introduction

Iron(II) sulphates are known to exist in several hydrated forms of which melanterite and rozenite are common in nature. They generally possess low crystal symmetries. The hyperfine and crystalline structures of these two iron(II) sulphates of interest in aqueous corrosion, are investigated. Melanterite sample of high purity is bought from PROLABO and rozenite is obtained by exposing melanterite to atmosphere.

2. Results and discussion

Melanterite, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, crystallizes in the space group $P 2_1/c$ with $a = 14.072 \text{ \AA}$, $b = 6.503 \text{ \AA}$, $c = 11.041 \text{ \AA}$ and $\beta = 105^\circ 34'$ [1]. The cell contains two different $\text{Fe}(\text{H}_2\text{O})_6^{2+}$ octahedra, one SO_4^{2-} tetrahedron and a seventh water molecule which is not coordinated to Fe^{2+} . Computer fittings of the Mössbauer spectra run at 78 K and 295 K (Figure 1) reveal the two sites. Fittings based only on one iron site [3-8] yield larger statistical errors. One notices that the two sites are equally populated (Table 1) and have different quadrupole splittings indicating that the two directions of the electric field gradient axes are also different. The total quadrupole splitting (QS) in ferrous compounds is given by : $\text{QS} = 1/2 e^2 q Q (1 + 1/3 \eta^2)^{1/2}$. The values of the field gradients q , taking $Q = 0.2 \text{ barn}$ [9] and $\eta = 0.1$ (2,10) are in Table 2. We may express q and η by :

$$q = V_{zz}/e = (1-R) q_{\text{val}} + (1-\gamma_\infty) q_{\text{lat}}$$

$$\text{and } \eta q = (V_{xx} - V_{yy})/e = (1-R) \eta_{\text{val}} q_{\text{val}} + (1-\gamma_\infty) \eta_{\text{lat}} q_{\text{lat}}$$

where q_{val} and q_{lat} are the contributions from the charge distribution of the non spherical valence electrons and from the lattice respectively, η_{val} and η_{lat} their asymmetry parameters, and $(1-R)$ and $(1-\gamma_\infty)$ their corresponding Sternheimer shielding factors.

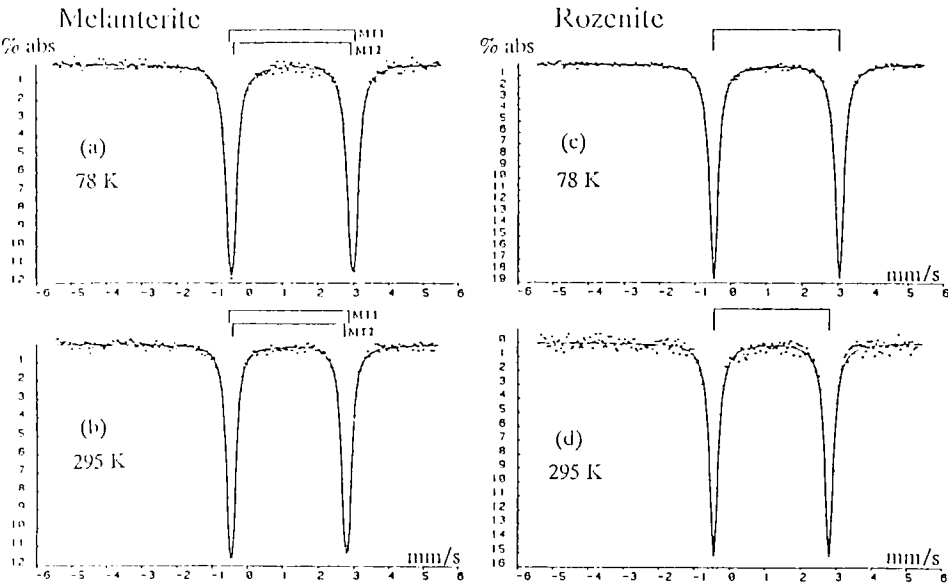


FIG. 1. Mössbauer spectra of Melanterite run at (a) 78 K and (b) 295 K and of Rozenite run at (c) 78 K and (d) 295 K.

Table 1
Mössbauer parameters of melanterite $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ and rozenite $\text{FeSO}_4 \cdot 4\text{H}_2\text{O}$.

Temperature	78 K		295 K	
Compound	Melanterite		Melanterite	Rozenite
IS (mm/s)	1.369	1.366	1.248	1.259
QS (mm/s)	3.551	3.286	3.373	3.256
Γ (mm/s)	0.286	0.286	0.249	0.285
%	50.1	49.9	49.8	50.2

IS = isomer shift with respect to a iron, QS = quadrupole splitting, Γ = line width.

The lattice contributions can be evaluated by assuming the neighbouring ions to be point charges and performing lattice summations [6, 9-11] for each site. By using the values of q_{lat} and η_{lat} calculated from the results of Nozik and Kaplan [9] and the best values of $(1-R) = 0.68$ and $(1-\gamma_{\infty}) = 12$ [6], the calculated values of q_{val} and η_{val} are given in Table 2. The reduction of the valence contribution to the QS for each site with temperature can be accounted for by the thermal populations of the energy levels. According to Ingalls [6], the

ground state wave orbital of Fe^{2+} in melanterite is d_{xy} ; the variation of QS_{val} with temperature can thus be written as (6) : $QS = (2/7) e^2 Q (1-R) \langle r^{-3} \rangle_0 \alpha^2 F(\Delta_1, \Delta_2, T)$ which with $\langle r^{-3} \rangle_0 = 4.8$ a.u., becomes $QS_{\text{val}} = 5.595 \alpha^2 F$. The function F accounts for the variations of the QS at different temperatures and α^2 is known as the covalency factor due to some charge transfer between the paramagnetic ion and the surrounding oxygen atoms. The effect of the spin-orbit interaction yields smaller contribution to F , and since it is only presumably important at low temperature, this effect can be neglected. The expression for F upon neglecting the spin-orbit effect is given by [6] :

$$F = Z^{-1} \{ 1 + \exp(-2\Delta_1/kT) + \exp(-2\Delta_2/kT) - \exp(-\Delta_1/kT) - \exp(-\Delta_2/kT) - \exp[-(\Delta_1 + \Delta_2)/kT] \}^{1/2}$$

where $Z = 1 + \exp(-\Delta_1/kT) + \exp(-\Delta_2/kT)$. $\Delta_1 = 480 \text{ cm}^{-1}$ and $\Delta_2 = 1300 \text{ cm}^{-1}$ are the separations of the $\{xy\}$ state from the $\{yz\}$ and $\{xz\}$ states respectively. If these separations are the same for the two sites in melanterite, then $QS_{\text{val}} = 5.594 \alpha^2$ and $4.864 \alpha^2$ at 78 and 295 K respectively. The values of α^2 , based on these assumptions and on the calculated values of QS_{val} are given in Table 2. No definite conclusion can be made on the difference in α^2 at each temperature since for example, if we base our analysis on the value of $Q \approx 0.3$ barn given by Ingalls [6], QS_{val} have the same magnitude corresponding to the average of the QS of the two sites at each temperature. However, since q is always positive for the two sites in melanterite [2], our observation is in good agreement with that of Nozik and Kaplan and disagrees with Ingalls' suggestion that QS_{val} and QS_{lat} are always of opposite signs.

Table 2

Field gradients q , asymmetry parameters η , lattice and valence contributions to the quadrupole splittings and the covalency factors for the two ferrous sites in melanterite.
 q values are given in 10^{24} cm^{-3} and QS in mm/s .

T (K)	Site	q	q_{val}	η_{val}	QS_{val}	q_{lat}	η_{lat}	QS_{lat}	α^2
78	1	11.818	16.989	0.084	3.470	+0.0221	0.82	+0.081	0.620
	2	10.936	16.490	0.112	3.371	-0.0231	0.59	-0.085	0.603
295	1	11.225	16.118	0.083	3.291	+0.0221	0.82	+0.082	0.677
	2	10.400	15.702	0.113	3.210	-0.0231	0.59	-0.085	0.660

Rozenite, $\text{FeSO}_4 \cdot 4\text{H}_2\text{O}$, exists naturally [12] or can be obtained by exposing melanterite to a relatively low humid atmosphere [13]. It crystallizes in the space group $P2_1/n$ [14] with $a = 5.97 \text{ \AA}$, $b = 13.64 \text{ \AA}$, $c = 7.97 \text{ \AA}$ and $\beta = 90^\circ 26'$. The iron ions are in octahedral sites connected to four water molecules and two oxygen atoms from different adjacent SO_4^{2-} ions

they all possess the same symmetry and the octahedra are less distorted than in melanterite. Mössbauer spectra of rozenite obtained by exposing melanterite to dry atmosphere for several weeks, show only one doublet with Mössbauer parameters close to those of the unique site previously given in the literature for melanterite. X-ray diffraction analysis of this sample indicates only the presence of rozenite.

Mössbauer spectrum, run at 295 K, of a melanterite sample partially transformed into rozenite displays at least five doublets ; two come from melanterite, one from rozenite and two other doublets with QS of 3.7 mm/s and 2.6 mm/s ; this fact stresses the importance of the lattice contributions to the value of QS. Furthermore, the maximum low temperature QS of Fe^{2+} ions in ionic crystals could be larger than 3.7 mm/s previously suggested by Ingalls [15].

3. Conclusions

Melanterite exhibits two quadrupole splittings while rozenite exhibits only one in conformity with their crystallography. The differences in the quadrupole splittings of the two sites of melanterite is partially due to lattice contributions, the significance of which is more noticeable from the sample of melanterite partially transformed into rozenite. The lattice contributions from the two sites are of the same magnitude, but of opposite signs in agreement with Nozik and Kaplan 's result [9].

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